

AptitudeMind — Today's Development Progress

Date: 21 September 2026

Focus: Question generation, deterministic answer verification, debugging, and GitHub synchronization.

1. Work Completed Today

Area	Result	Details
Answer Engine	Completed	Supports Percentage, Profit, Loss, Average, Simple Interest, Distance, Speed, Time, Linear Equation, and
Answer Engine Tests	13/13 passed	All valid-answer and expected-error test cases passed.
Generator	Fixed	Changed verification check from <code>verification['success']</code> to <code>verification['status'] == 'success'</code> .
Generator Prompt	Improved	Strengthened Algebra rules for integer coefficients and simple real roots.
Generator Parsing	Improved	Added structural and parameter checks for Algebra questions.
Verification Flow	Working	Generated questions pass Validator and then deterministic Answer Engine verification.
Successful Test	Passed	Generated $2x^2 + 5x - 3 = 0$ with options including $1/2$; Validator and Answer Engine accepted it.
GitHub	Synced	Local commit was rebased onto origin/main and successfully pushed.

2. Debugging Lessons

- The Answer Engine contract uses a status field rather than a boolean success field.
- Raw JSON braces inside an f-string caused a formatting error; the final prompt uses placeholder replacement instead.
- The LLM generates questions, while deterministic Python validation and the Answer Engine verify mathematical correctness.
- Invalid quadratic generation was traced to generation quality, then addressed with stronger prompts and deterministic pre-checks.

3. Final Generator Verification

Question: Solve for x: $2x^2 + 5x - 3 = 0$

Options: 1. $3/2$ 2. $1/2$ 3. 2 4. $-3/2$

Topic: Algebra **Type:** QUADRATIC_EQUATION **Difficulty:** Hard **Company:** TCS

Parameters: a=2, b=5, c=-3

Result: Validator passed and Answer Engine verification passed.

4. GitHub Status

The main branch is now up to date with origin/main, confirming the push succeeded.

agent.py and answer_engine.py still have unstaged local modifications and were intentionally not committed in this push.

FETCH_HEAD was a Git-generated untracked file and should not be added to the repository.

5. Current AptitudeMind Flow

Agent → Generator → JSON Parsing → Validator → Answer Engine → Evaluator → Memory → Adaptive Difficulty

Today's work strengthened the Generator → Validator → Answer Engine section and synchronized the generator changes with GitHub.

6. Next Step

Test the complete Agent → Generator → Validator → Answer Engine → Evaluator → Memory flow. Keep generator.py unchanged unless integration testing reveals a new defect.